

Integer Linear Forms for a Factorial Reciprocal Series

Cancellation and attainable moments for Erdős Problem #68

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ABSTRACT

We classify the finite integer vectors that cancel the first weighted summands of a factorial reciprocal series. An integral basis turns these cancellations into prescribed coordinates; excluding index one leaves a single Diophantine equation. We determine its attainable factorial moments by a gcd taken over fewer than $2D^2$ indices, where D is the cancellation depth. A worked construction attains the least positive moment 1380 at depth four. The additional nonintegrality condition required for an irrationality proof remains open.

1 Introduction

Erdős asked whether replacing $n!$ by $n! - 1$ preserves the irrationality of the reciprocal sum [1, p. 102]. The question appears as Problem 68 in Bloom’s catalogue [2].

Problem 1.1 (Erdős Problem #68). Is

$$S = \sum_{n \geq 2} \frac{1}{n! - 1}$$

irrational?

The elementary proof that e is irrational uses two properties of factorials: they clear every earlier denominator, and they leave a small positive tail. The first property fails here. If $H_m = \sum_{n=2}^m (n! - 1)^{-1}$, then already $3!H_3 = 36/5$. We shall instead construct linear forms $MS + k$, with $M, k \in \mathbb{Z}$, in which several initial summands cancel. A non-integral form excludes every rational value $S = a/q$ for which $q \mid M$.

The cancellation problem admits an exact solution. We give an integral basis in which each basis vector changes just one weighted sum (Theorem 2.1). Thus cancellation amounts to prescribing coordinates. The basis temporarily allows an index-one coefficient; removing it leaves one linear Diophantine equation. Its solvability determines all possible factorial moments, and Theorem 3.1 computes the resulting ideal by a finite gcd.

**Authorship and AI use.* Will Cook built and directed the research infrastructure and reviewed the claims when he could. AI agents did most of the research and drafting. Cook did not independently verify every claim.

The construction begins with $n(n-1)! - n! = 0$. For our weights, this adjacent difference is still zero except when a floor exponent changes, which happens at divisors of n . Correcting the proper divisors in increasing order isolates the weighted sum at n . The corrections are integral and triangular, so they give every integer solution, with no search for suitable coefficients.

At depth four the least positive moment is 1380. We derive an attaining vector and place its remainder between consecutive integers. The algebraic classification holds independently of whether S is rational. To prove irrationality, one would need a nonintegral form with $q \mid M$ for each prescribed $q > 0$. Primitive vectors whose moments eventually have this divisibility are available, but nonintegrality of their remainders remains unproved.

Hančl and Tijdeman's tail-integrality lemma treats factorial series with integer coefficients [4, Lemma 2.1 and the following remark, p. 385]. Their factorial-scaled partial sums are integers. Our prefixes lack this property, so the integer-difference identity below replaces the clearing step. Koepf and Schmiersau's factorial-digit criterion [3, Example 3.2, p. 121] is used in the [companion's digit argument](#) to give an equivalent eventual-pattern condition for $S - e + 2$. Failure of that pattern at arbitrarily large indices also remains unproved.

Sections 2 and 3 give the coordinate construction and finite gcd. Section 4 uses them to construct and test a minimal-moment vector. Section 5 states the remaining nonintegrality question, with the secondary arithmetic criteria developed in the companion.

2 An integral basis

Modulo $d! - 1$, a factor $d!$ can be removed from a numerator without changing its residue. We use this freedom to replace $n!$ by a smaller integer whose change from $n-1$ to n is easy to follow. For $d \geq 2$ and $n \geq 1$, put

$$W_{d,n} = \frac{n!}{(d!)^{\lfloor n/d \rfloor}}.$$

Writing $n = kd + r$, $0 \leq r < d$, shows that $W_{d,n}$ is $r!$ times the multinomial coefficient $n!/((d!)^k r!)$. In particular it is an integer, and $n! = (d!)^k W_{d,n}$ gives $W_{d,n} \equiv n! \pmod{d! - 1}$. For example, $W_{3,4} = 4$ and $(4! - W_{3,4})/(3! - 1) = 4$. The exponent counts complete blocks of d indices. It is this choice that makes its jumps occur exactly at multiples of d .

For a finitely supported integer vector $\lambda = (\lambda_n)_{n \geq 1}$, we define its factorial moment and weighted sums by

$$M(\lambda) = \sum_n \lambda_n n!, \quad V_d(\lambda) = \sum_n \lambda_n W_{d,n},$$

and let

$$\mathcal{R}(\lambda) = \sum_{d \geq 2} \frac{V_d(\lambda)}{d! - 1}.$$

Choose $N \geq 2$ beyond the support of λ . For $d > N$ the floor exponents are zero, so $V_d = M$ and the series converges absolutely. Only the first $N-1$ terms can differ from

those of MS , and the weight congruence gives

$$\mathcal{R}(\lambda) - M(\lambda)S = \sum_{d=2}^N \frac{V_d(\lambda) - M(\lambda)}{d! - 1} \in \mathbb{Z}. \quad (1)$$

We now solve $V_2 = \dots = V_D = 0$ in integer vectors, allowing either sign of M . Write e_n for the unit vector at index n . The auxiliary coordinate at index one will make the basis triangular. It occurs only in the coefficient vector: the series still starts at $d = 2$, so no term $1/(1! - 1)$ is introduced. We impose $\lambda_1 = 0$ in Section 3.

The identity $n(n-1)! - n! = 0$ suggests starting with $T_n = ne_{n-1} - e_n$. At $n = 2$, the vector $U_2 = 2e_1 - e_2$ has $V_2 = 1$ and every other weighted sum zero. At $n = 4$, the difference $T_4 = 4e_3 - e_4$ has two nonzero weighted sums. One correction suffices:

Vector	M	V_2	V_3	V_4
T_4	0	6	0	23
$6U_2$	0	6	0	0
$T_4 - 6U_2$	0	0	0	23

All later weighted sums are zero because the moment is zero. The last row defines U_4 : it isolates $V_4 = 4! - 1$ without changing the moment. For a general n , the unwanted sums are precisely those at proper divisors of n , and their correcting vectors have already been built.

Theorem 2.1 (an integer basis with prescribed weighted sums). [Lean](#) · [Comparator](#) *Set*

[Lean](#)
[Comparator](#)

$$T_n = ne_{n-1} - e_n, \quad U_n = T_n - \sum_{\substack{d|n \\ 2 \leq d < n}} W_{d,n} U_d \quad (n \geq 2).$$

Then

$$M(U_n) = 0, \quad V_d(U_n) = (d! - 1)\mathbf{1}_{d=n}.$$

The vectors $e_1, U_2, U_3, \dots$ form an integral basis. Every finite vector has the unique finite expansion

$$\lambda = M(\lambda)e_1 + \sum_{d \geq 2} \frac{V_d(\lambda) - M(\lambda)}{d! - 1} U_d. \quad (2)$$

Proof. The moment of T_n is zero. If $d \nmid n$, the floor exponents at $n-1$ and n agree, so $nW_{d,n-1} = W_{d,n}$. If $d \mid n$, the exponent increases by one, so $nW_{d,n-1} = d!W_{d,n}$. Hence

$$V_d(T_n) = (d! - 1)W_{d,n}\mathbf{1}_{d|n}.$$

Suppose the asserted identities hold below n . For each proper divisor d , subtracting $W_{d,n}U_d$ cancels the sum at d and changes none of the others. The sum at n remains $n! - 1$, since $W_{n,n} = 1$. This proves the identities by induction.

To see that no integer vectors are missed, restrict to indices $1, \dots, N$. The columns $e_1, T_2, \dots, T_N$ form a triangular integer matrix with diagonal $1, -1, \dots, -1$. The correction from T_n to U_n uses only earlier columns and leaves its diagonal entry unchanged. This

second change is triangular with diagonal one. Both changes therefore have integer inverses. Equivalently, one can recover the coefficients by back-substitution, dividing only by 1 or -1 .

In the resulting expansion, applying M gives the coefficient of e_1 . If the coefficient of U_d is c_d , applying V_d gives $V_d = M + (d! - 1)c_d$, which proves (2). For $d > N$ that coefficient is zero because $V_d = M$. $\square$

For a prime $p \geq 3$ the divisor sum is empty, and $U_p = pe_{p-1} - e_p$. At a composite index the correction terms remove the extra weighted sums. For example,

$$U_9 = 9e_8 - e_9 - 5040e_2 + 1680e_3. \quad (3)$$

Each U_n has moment zero and remainder one. We can thus alter an individual weighted sum by adding a multiple of U_n , while the fractional part of the remainder stays fixed. This will separate the cancellation equations from the eventual nonintegrality question.

3 Attainable moments

Fix a cancellation depth $D \geq 2$. The basis separates the equations $V_2 = \dots = V_D = 0$ from the restriction $\lambda_1 = 0$. We first solve the equations with index one still allowed. The weight congruence, or the coordinates in (2), gives

$$V_d(\lambda) \equiv M(\lambda) \pmod{d! - 1}. \quad (4)$$

Cancellation through D therefore forces the moment to be divisible by

$$L_D = \text{lcm}_{2 \leq d \leq D}(d! - 1).$$

This necessary condition is sufficient while index one is allowed: L_De_1 has every weighted sum equal to L_D , and subtracting $L_D/(d! - 1)$ copies of U_d sets the d th sum to zero. Thus put

$$K_D = L_De_1 - \sum_{d=2}^D \frac{L_D}{d! - 1} U_d.$$

Uniqueness of the coordinates now gives the complete solution

$$V_2 = \dots = V_D = 0 \iff \lambda = tK_D + \sum_{n>D} z_n U_n, \quad M = tL_D, \quad (5)$$

with $t \in \mathbb{Z}$ and finitely many nonzero integers z_n . The lower coordinates are fixed by t ; all higher coordinates remain free. Since $\mathcal{R}(e_1) = S$ and $\mathcal{R}(U_n) = 1$, we also have

$$\mathcal{R}\left(tK_D + \sum_{n>D} z_n U_n\right) = tL_D(S - H_D) + \sum_{n>D} z_n, \quad H_D = \sum_{d=2}^D \frac{1}{d! - 1}. \quad (6)$$

It remains to remove index one. Write a_D for its coefficient in K_D , and u_n for its coefficient in U_n . All other coordinates already lie in the permitted range $n \geq 2$, so the only remaining equation is

$$ta_D + \sum_{n>D} z_n u_n = 0. \quad (7)$$

The higher corrections can change this coefficient by exactly the integer multiples of $g_D = \gcd\{u_n : n > D\}$. Thus the equation is soluble precisely when $g_D \mid ta_D$. For example, at depth four we shall find $a_4 = -55$ and $g_4 = 60$. Removing index one then requires $60 \mid 55t$, or $12 \mid t$; the moment $115t$ must be a multiple of 1380.

The same calculation in general shows that the attainable moments are

$$\mu_D \mathbb{Z}, \quad \mu_D = L_D \frac{g_D}{\gcd(g_D, a_D)}. \quad (8)$$

Indeed, division by $\gcd(g_D, a_D)$ leaves two coprime integers, so $g_D \mid ta_D$ holds exactly when $g_D / \gcd(g_D, a_D) \mid t$. Bézout's identity supplies the required finite combination of the u_n . The next theorem proves $g_D > 0$ and bounds the indices needed for this combination. A vector attaining the least positive moment μ_D is automatically primitive: a common factor of its coefficients would give a smaller positive moment after division.

3.1 A finite gcd calculation

The recursion for the first coordinate is

$$u_2 = 2, \quad u_n = - \sum_{\substack{d \mid n \\ 2 \leq d < n}} W_{d,n} u_d \quad (n > 2).$$

Induction gives $u_n = 0$ at odd indices. At twice a prime it gives $u_{2p} = -(2p)!/2^{p-1} \neq 0$, including $p = 2$, so the tail contains a nonzero coefficient and $g_D > 0$. To compute this infinite gcd, we show that a specified finite interval already forces divisibility of every later coefficient.

A finite calculation can miss a later drop in the gcd. Here the recurrence prevents such a drop beyond a controlled index. Large proper divisors inherit divisibility from earlier coefficients; for the remaining small divisors, the weights themselves supply it.

Theorem 3.1 (a finite formula for the gcd). [Lean](#) · [Comparator](#) Choose a prime ℓ with $D/2 < \ell \leq D$ and put $H = D(2\ell - 1)$. Then

[Lean](#)
[Comparator](#)

$$g_D = \gcd(u_{D+1}, \dots, u_H), \quad H < 2D^2.$$

Proof. Let $g = \gcd(u_{D+1}, \dots, u_H)$. The index 2ℓ belongs to this interval, and $u_{2\ell} = -(2\ell)!/2^{\ell-1} \neq 0$. Consequently $g > 0$ and $g \mid (2\ell)!$.

We prove $g \mid u_n$ for every $n > D$ by strong induction. This is true up to H by definition. For $n > H$, consider a proper divisor $d \geq 2$ in the recurrence for u_n . If $d > D$, the induction hypothesis gives $g \mid u_d$. If $d \leq D$, put $k = n/d$. The choice $H = D(2\ell - 1)$ ensures $k \geq 2\ell$. Moreover,

$$\frac{W_{d,n}}{k!} = \frac{n!}{(d!)^k k!} \in \mathbb{Z} :$$

this quotient counts partitions of n labelled objects into k unordered blocks of size d . Hence $(2\ell)! \mid k! \mid W_{d,n}$, so $g \mid W_{d,n}$ in the small-divisor case as well. Every product $W_{d,n}u_d$ in the recurrence is therefore divisible by g .

Thus g divides the whole tail. Conversely, every common divisor of the tail divides its finite subfamily $u_{D+1}, \dots, u_H$. The two gcds are equal, and $H \leq D(2D - 1) < 2D^2$. $\square$

Bertrand's postulate supplies the prime ℓ when $D \geq 3$; for $D = 2$ take $\ell = 2$. The proof requires $n/d \geq 2\ell$ for every small divisor, which is why it uses the horizon $H = D(2\ell - 1)$. A truncation at $2D$ is not justified by this argument.

The finite gcd also gives a construction. Compute integers b_n with $\sum_{n=D+1}^H b_n u_n = g_D$ by the extended Euclidean algorithm, and put $h = \gcd(g_D, a_D)$. Taking

$$t = g_D/h, \quad z_n = -(a_D/h)b_n \quad (D < n \leq H)$$

makes $ta_D + \sum z_n u_n = 0$. Substitution in (5) therefore produces a vector with moment μ_D . The same finite calculation proves minimality and supplies an attaining vector. Additional support or coefficient-norm requirements are separate optimisation problems.

4 A vector of least positive moment

At depth four the recursion gives

$$L_4 = 115, \quad K_4 = -55e_1 + 16e_2 + 3e_3 + 5e_4, \quad a_4 = -55.$$

Take $\ell = 3$ in Theorem 3.1, so that $H = 20$. The coefficients in this finite interval have gcd 60. There is a useful short proof: $u_6 = -180$, $u_8 = -4200$ and $23u_6 - u_8 = 60$, while

$$\text{lcm}(1, \dots, n) \mid u_n \tag{9}$$

gives the reverse divisibility for every $n > 4$. To prove (9), put $\Lambda_n = \text{lcm}(1, \dots, n)$. For a prime r , write v_r for its valuation. If $d \mid n$, Legendre's formula gives

$$v_r(W_{d,n}) = \sum_{j \geq 1} \left(\left\lfloor \frac{n}{r^j} \right\rfloor - \frac{n}{d} \left\lfloor \frac{d}{r^j} \right\rfloor \right) \geq v_r(\Lambda_n) - v_r(\Lambda_d).$$

Each summand is nonnegative, and each power $d < r^j \leq n$ contributes at least one. Hence $\Lambda_n \mid W_{d,n}\Lambda_d$, and induction in the recurrence, starting from $u_2 = 2$, proves the assertion. Formula (8) now gives

$$\mu_4 = 115 \frac{60}{\gcd(60, 55)} = 1380.$$

We construct a vector with this moment by taking $t = 12$ in (7). The choice $z_6 = -27$, $z_8 = 1$ works because

$$12(-55) - 27(-180) - 4200 = 0.$$

The vector $12K_4 - 27U_6 + U_8$ therefore has no index-one coefficient and cancels V_2, V_3, V_4 . To remove its coefficients at indices four and seven, add $-26U_5 + 8U_7$. These corrections still cancel through four and do not restore index one, since $u_5 = u_7 = 0$. Thus

$$\lambda = 12K_4 - 26U_5 - 27U_6 + 8U_7 + U_8,$$

which expands to

$$\lambda = 1482e_2 - 784e_3 - 136e_5 + 83e_6 - e_8, \quad (M, V_2, V_3, V_4) = (1380, 0, 0, 0). \tag{10}$$

n	λ_n	$n!$	$W_{2,n}$	$W_{3,n}$	$W_{4,n}$
2	1482	2	1	2	2
3	-784	6	3	1	6
5	-136	120	30	20	5
6	83	720	90	20	30
8	-1	40320	2520	1120	70
Weighted totals		1380	0	0	0

Table 1: The depth-four vector and its checks. To obtain the last row, multiply each of the final four columns by λ_n before summing.

Table 1 verifies all four sums directly.

The moment and all cancellations follow from the coordinates before expansion. Moreover, the integer correction to the remainder in (6) is the sum $-26-27+8+1 = -44$. Thus

$$\mathcal{R}(\lambda) = 1380(S - H_4) - 44.$$

To test this remainder, we use a tail estimate. $(n+1)! - 1 > (n+1)(n! - 1)$ gives the geometric tail bound

$$0 < S - H_N < \frac{N+2}{(N+1)((N+1)! - 1)} < \frac{2}{(N+1)! - 1} \quad (N \geq 2). \quad (11)$$

The majorant starts with $1/((N+1)! - 1)$ and has ratio $1/(N+2)$.

Exact rational arithmetic gives

$$-31 + \frac{4}{5} < 1380 \sum_{d=5}^8 \frac{1}{d! - 1} - 44 < -31 + \frac{5}{6}, \quad \frac{2 \cdot 1380}{9! - 1} < \frac{1}{100}.$$

The positive remaining tail is less than $2/(9! - 1)$, so $-31 < \mathcal{R}(\lambda) < -30$. If $S = a/q$ with $q \mid 1380$, (1) would make this remainder an integer. Every divisor of 1380 is therefore excluded.

The companion gives the [alternative dual congruence, support and coefficient-norm refinements](#). They concern different optimisation questions. Minimality of the moment here follows from the ideal formula, without requiring a sparsest vector.

5 Nonintegrality of the remainder

Fix a vector with $V_2 = \dots = V_D = 0$ and $M > 0$. We may choose the cutoff $N \geq D$ anywhere at or beyond its support. Separate the finite signed sum from the positive tail:

$$A_N = \sum_{d=D+1}^N \frac{V_d(\lambda)}{d! - 1}, \quad \mathcal{R}(\lambda) = A_N + M(S - H_N).$$

By (11), $0 < S - H_N < 2/((N + 1)! - 1)$. We can prove nonintegrality by comparing its upper bound with the distance from A_N to the next integer:

$$\frac{2M}{(N + 1)! - 1} < \lfloor A_N \rfloor + 1 - A_N. \quad (12)$$

Under this inequality, $\mathcal{R}(\lambda)$ lies strictly between consecutive integers, excluding every denominator dividing M . The argument allows either sign of A_N . Its strict gap is one when A_N is integral and can be arbitrarily small when A_N approaches an integer from below.

There are primitive solutions with $0 < M < N!$ and moments eventually divisible by any prescribed q . The [progression construction in the companion](#) gives such a family by turning $V_d = 0$ into roots of a polynomial. At its support endpoint the omitted tail is less than $2/N$, but the gap on the right of (12) may shrink too. Irrationality requires the comparison for a vector whose moment is divisible by each prescribed q , not merely an unbounded sequence of moments. No minimal-support assertion is made for this family.

Increasing the cutoff is a different operation from choosing a new vector. For $-6e_2 + e_4$ the support ends at four, where the test fails because $9/115 < 24/119$. At $N = 5$ it succeeds because $13376/13685 > 24/719$. More generally, for a fixed vector with $M > 0$, the quantities A_N increase to $\mathcal{R}$. If this limit is nonintegral, their gaps to the next integer tend to a positive number, so the test eventually succeeds. If the limit is integral, the eventual gap equals the omitted tail and the test fails. Thus extending the cutoff eventually detects a nonintegral remainder, but leaves both the remainder and the moment unchanged.

Moment-zero corrections also have a precise limitation. Each U_n changes the remainder by one. More generally, (1) shows that two vectors with the same moment have remainders differing by an integer. Such changes can simplify support and weighted sums, as they did in the example, but cannot change whether the remainder is integral.

The companion treats the [equivalent carry condition](#) and the [complementary-denominator tail inequality](#). It also records two larger finite computations: a non-unit carry at $m = 300000$ excludes every $q \mid 299999!$, and a continued-fraction enclosure excludes every $q < 2^{39990}$. Both computations were performed outside Lean; their [algorithms, exact outputs and scope](#) are given there. The factorial-divisibility exclusion does not exclude all denominators with only small prime factors, whose exponents may be larger. Neither finite restriction establishes the required condition at arbitrarily large indices.

6 Verification

Formal proofs. A result with a kernel-checked Lean proof carries links beside its heading and in the margin; both open the same evidence. *Lean* opens the proof: the declaration itself when one declaration states the whole result, otherwise the list of declarations that together state it. Inline citations to individual declarations elsewhere may identify ingredients of a proof. *Comparator* opens the recorded comparison, in which the same statement, written again from Mathlib alone in a separate repository, was compared

with our proof by Lean’s Comparator tool, allowing only the three standard axioms. *Comparator pending* means that the complete statement has no recorded comparison yet. A dagger on the Lean mark means that the Lean proof assumes an input named just below the result. A result without a mark has no Lean proof of its whole statement; what is checked is said below it. The [evidence record](#) gives every declaration, version and check. Comparator checks statements, axioms and kernel acceptance; it does not establish novelty or peer review. These checks show that the stated propositions are proved; whether they are the right propositions is a question the reader can settle by comparing them with the text.

The integral-coordinate theorem and finite-horizon gcd formula have recorded exact Lean bindings. The [companion source concordance](#) identifies their declarations and the attainable-moment theorem. Its large finite computations are separate from these kernel proofs.

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References

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- [4] Jaroslav Hančl and Robert Tijdeman. [On the irrationality of factorial series](#). *Acta Arithmetica* **118** (4) (2005), 383–401.